

Abhijeet Dhali

Computer Science Graduate Student | Technical Support | Software & Research Experience | Ontario, Canada
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PROFESSIONAL SUMMARY

Motivated and adaptable computer science professional with experience across **technical support, software development, teaching, and applied research**. Strong background in **Python programming, troubleshooting, machine learning, Linux/Windows environments, and technical communication**. Experienced in supporting users, building and testing software pipelines, conducting analysis, and collaborating in academic and professional settings. Seeking general opportunities across **technology, IT support, software, data, and related roles**.

PROFESSIONAL EXPERIENCE

Mitacs Accelerate Research Intern, Brock University

May 2026 – Present

- Researching and developing RapidBOM, a privacy-preserving AI system designed for automated Bill of Materials (BOM) extraction and sourcing from engineering documents.
- Designing machine learning workflows to securely process and extract structured data from complex engineering files.
- Collaborating with faculty supervisors and industry partners to translate research objectives into a deployable, privacy-centric AI solution.

Research Assistant, Brock University, Computer Science Department

Sep 2023 – Present

- Implemented and evaluated machine learning models using **Python, PyTorch Geometric**, and **NetworkX** for graph-based research projects.
- Designed experimental pipelines for model training, benchmarking, testing, and reproducibility in Linux-based environments.
- Conducted debugging, performance analysis, and optimization to improve reliability and accuracy of research systems.
- Collaborated with faculty and researchers to validate experiments, document findings, and present technical results.
- Managed datasets, configurations, and scripts for multiple concurrent projects in applied AI and secure systems research.

Technical Support Advisor, Concentrix Canada

Mar 2026 – May 2026

- Deliver high-quality technical support in a dynamic, high-volume environment by diagnosing and resolving a wide range of user issues, including system access, software functionality, and service disruptions.
- Analyze customer-reported problems to identify root causes and apply structured troubleshooting methodologies to achieve efficient and accurate resolutions.
- Utilize internal tools, knowledge bases, and workflows to investigate issues, ensuring adherence to operational and security protocols.
- Maintain detailed and accurate documentation of incidents, actions taken, and outcomes to support knowledge sharing and continuous process improvement.
- Escalate complex or unresolved issues with clear technical context, enabling faster resolution by specialized teams.
- Demonstrate strong communication and problem-solving skills by translating technical concepts into user-friendly guidance and ensuring a positive customer experience.

Teaching Assistant, Brock University, Computer Science Department

Sep 2023 – Present

- Delivered lab sessions and tutorials for undergraduate courses in programming, machine learning, and data structures.
- Assisted students with coding, debugging, and understanding core computer science concepts in Python-based coursework.
- Supported course delivery by preparing examples, verifying assignments, and helping maintain smooth lab operations.
- Communicated technical concepts clearly to diverse learners and provided one-on-one academic support when needed.

PROJECTS

SPROUT-GNN: Intrinsic Defenses Against Backdoor Attacks in High-Order GNNs

[GitHub]

- Developed and evaluated **Graph Neural Networks (GNN)**-based machine learning workflows using **Python**, **PyTorch Geometric**, and **NetworkX**.
- Applied extensive **debugging, testing, and pipeline optimization** to ensure reproducibility and stable model performance.
- Designed defense strategies that consistently reduced **attack success rate (ASR)** while maintaining high model accuracy across multiple datasets.

SAGNN+CS: A Resilient Graph Neural Network Framework Against Backdoor Threats

[GitHub]

- Implemented multiple GNN architectures (GCN, GAT, GraphSAGE, ESAN, SUN, SAGNN) using **Python** and **PyTorch**.
- Conducted performance benchmarking and robustness testing using **custom scripts** and automated experimental pipelines.
- Enhanced model resilience by achieving a strong balance between low **attack success rate (ASR)** and high predictive accuracy under adversarial conditions.

How Robust Are Higher-Order Graph Neural Networks to Backdoor Attacks?

[GitHub]

- Evaluated and compared higher-order GNNs (ESAN, SUN, SAGNN) under diverse adversarial attack scenarios.
- Identified trade-offs between model robustness and accuracy, highlighting the importance of balancing security and performance.
- Managed **experimental configurations, dataset preprocessing**, and automated evaluation using Python-based workflows.

The Power of Many: Investigating Defense Mechanisms for Resilient GNNs

[GitHub]

- Explored multiple **attack and defense mechanisms** (SBA-Samp, GTA, DPGBA) in controlled experimental environments.
- Improved model robustness by applying defense strategies that maintain performance while reducing vulnerability to attacks.
- Used structured logging and result tracking to ensure consistent and reliable evaluation of model behavior.

PUBLICATIONS

- **Intrinsic Defenses Against Backdoor Attacks in High-Order Graph Neural Networks via Semantic and Outlier-Guided Subgraph Policies (SPROUT-GNN).** *IEEE CASCON 2025*. [Link]
- **SAGNN+CS: A Resilient Graph Neural Network Framework Against Backdoor Threats.** *Canadian AI 2025*. [Link]
- **How Robust Are Higher-Order Graph Neural Networks to Backdoor Attacks?** *Canadian AI 2025*. [Link]
- **The Power of Many: Investigating Defense Mechanisms for Resilient Graph Neural Networks.** *IEEE Big Data Workshop 2024*. [Link]

EDUCATION

Master of Science in Computer Science, Brock University

Sep 2023 – Sep 2025

- Focus: **Machine Learning, Graph Neural Networks, Adversarial Robustness, and Scalable AI Systems**
- Average: 95%

SKILLS

Programming / Scripting	Python, Bash, C++, JavaScript, SQL
Operating Systems	Linux, Windows
Tools & Platforms	Git, Docker, VS Code, Jupyter, Anaconda, Streamlit
Technical Support	Troubleshooting, issue resolution, ticket documentation, customer communication, escalation handling
Networking & Systems	TCP/IP, REST APIs, virtualization, system-level debugging, basic networking concepts
Databases	MySQL, SQLite
Frameworks / Libraries	PyTorch, PyTorch Geometric, Scikit-learn
Work Practices	Agile collaboration, testing, debugging, reproducibility, technical documentation
Soft Skills	Problem Solving, Communication, Collaboration, Adaptability, Time Management

REFERENCE

Available upon request.